

Data Analysis Report

Patient Healthcare Analysis



(+92) 3440697283



muhammadsarramali@gmail.com



sarram-ali.github.io

Muhammad Sarram

18/June/2026



COMPANY INTRODUCTION

Our Company **MediCore Hospitals & Clinics** are a growing chain of multi-specialty hospitals operating in major cities. We treat thousands of patients every year across various departments (Cardiology, Pulmonology, Rheumatology, ENT, General Medicine, etc.).

Our FY 2024 data shows increasing patient volume, but we are facing challenges in:

- Rising treatment costs
- High repeat visits for chronic diseases
- Inefficient medicine inventory
- Uneven performance across specialties and severity levels

The management wants clear insights to improve patient care quality, reduce costs, and optimize resource allocation.

PROBLEM

- **Chronic diseases** dominate our patient load, but we don't know which categories are most expensive or have the highest repeat visits.
- Severe cases are driving up medicine costs and doctor workload.
- We need to understand patterns by **age, gender, insurance, disease severity, and doctor specialty**.
- **Goal:** Reduce unnecessary repeat visits, optimize medicine stocking, and improve treatment efficiency for high-severity patients.

KEY BUSINESS QUESTIONS

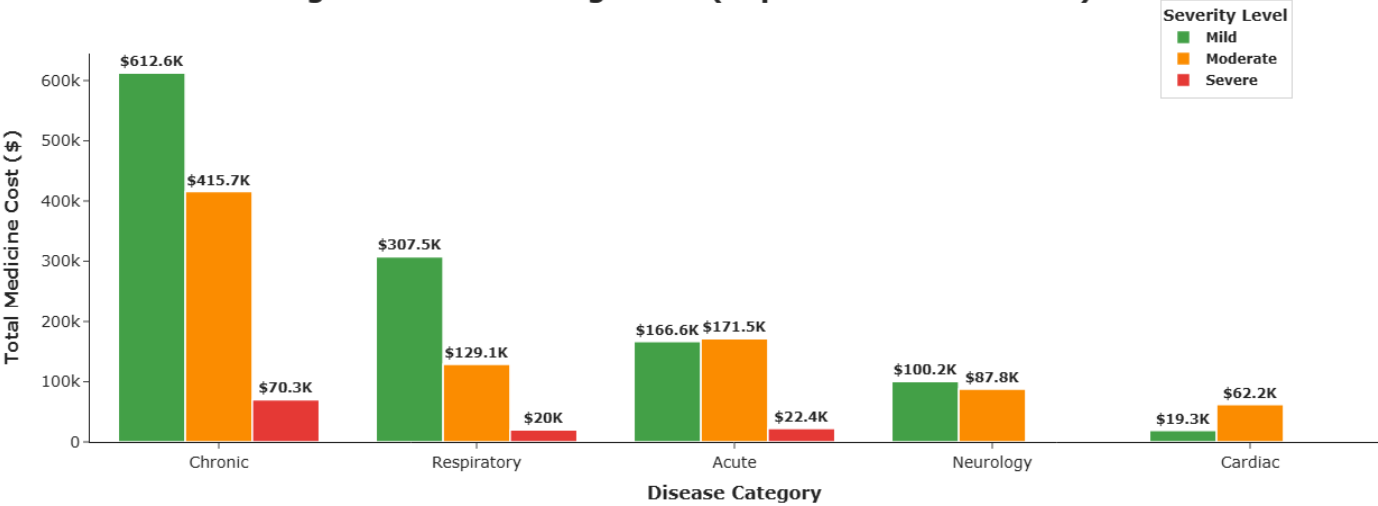
1. Revenue & Cost Signals
2. Patient Demographics & Distribution
3. Disease & Severity Analysis
4. Treatment & Cost Insights
5. Doctor & Hospital Performance
6. Visit Patterns & Efficiency

KEY FINDINGS

REVENUE & COST SIGNALS

- Total Medicine Cost: \$9,216,626

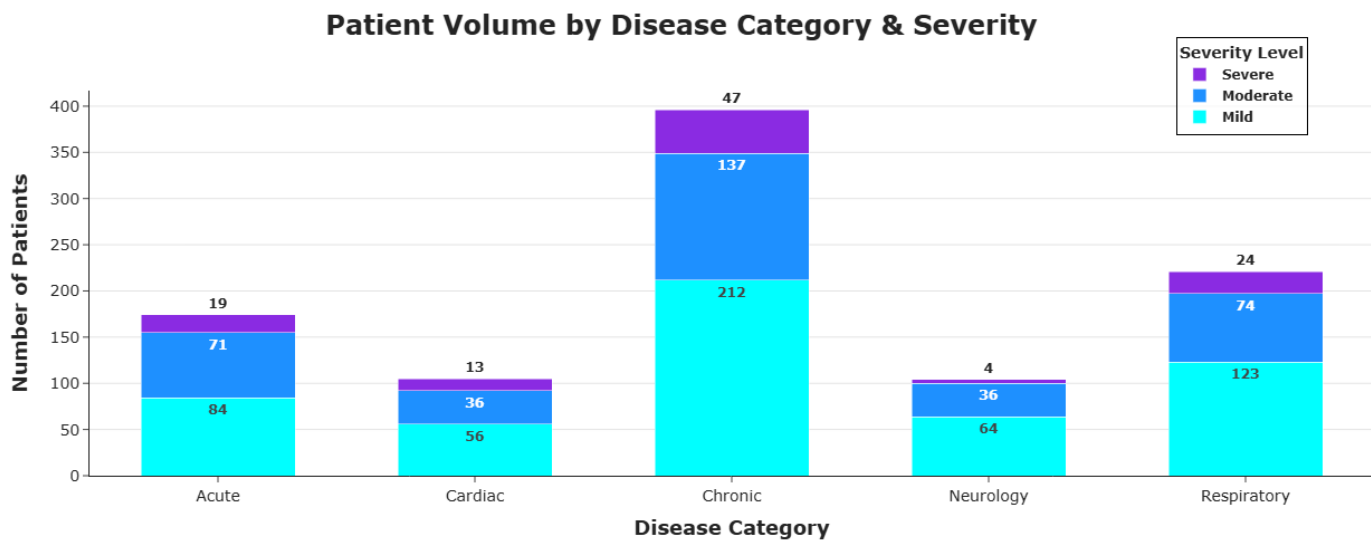
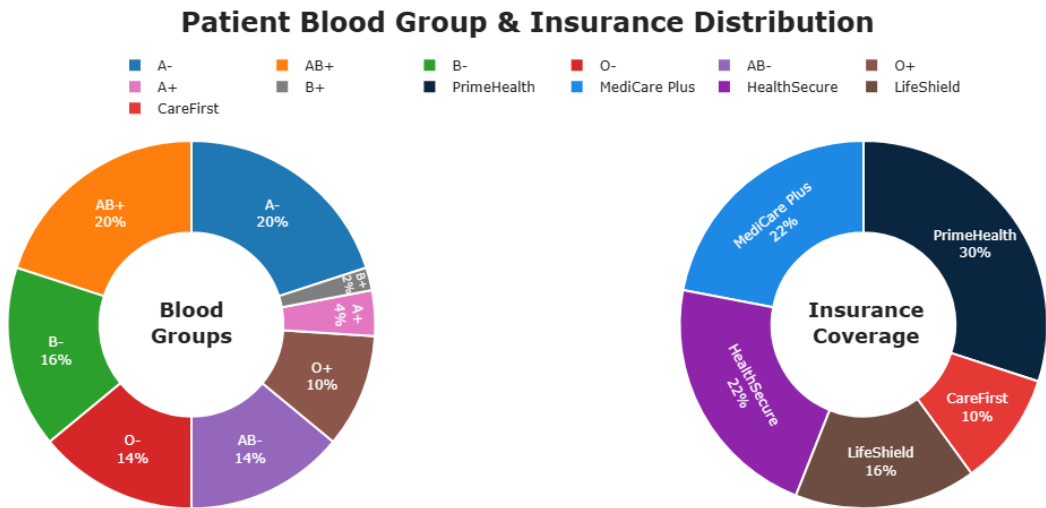
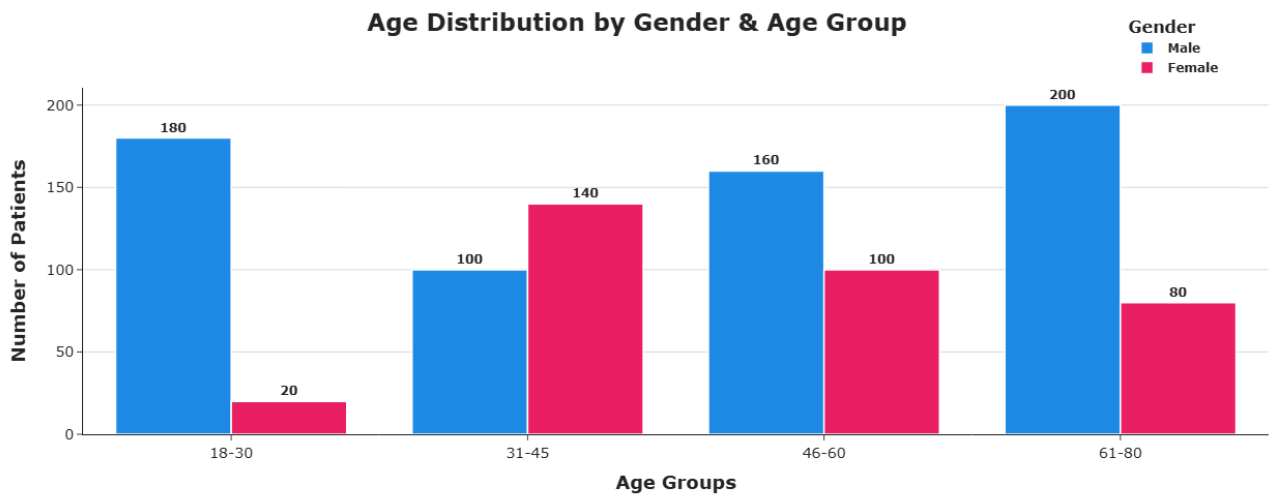
High-Cost Patient Segments (Top 10% Medicine Cost)



- Correlation (Age, Quantity, Cost_price, Visits, Total_Medicine_Cost)

	age	quantity	cost_price	number_of_visits	total_medicine_cost
age	1.00	-0.03	0.04	0.05	0.02
quantity	-0.03	1.00	0.00	0.03	0.64
cost_price	0.04	0.00	1.00	-0.08	0.70
number_of_visits	0.05	0.03	-0.08	1.00	-0.04
total_medicine_cost	0.02	0.64	0.70	-0.04	1.00

PATIENT DEMOGRAPHICS & DISTRIBUTION

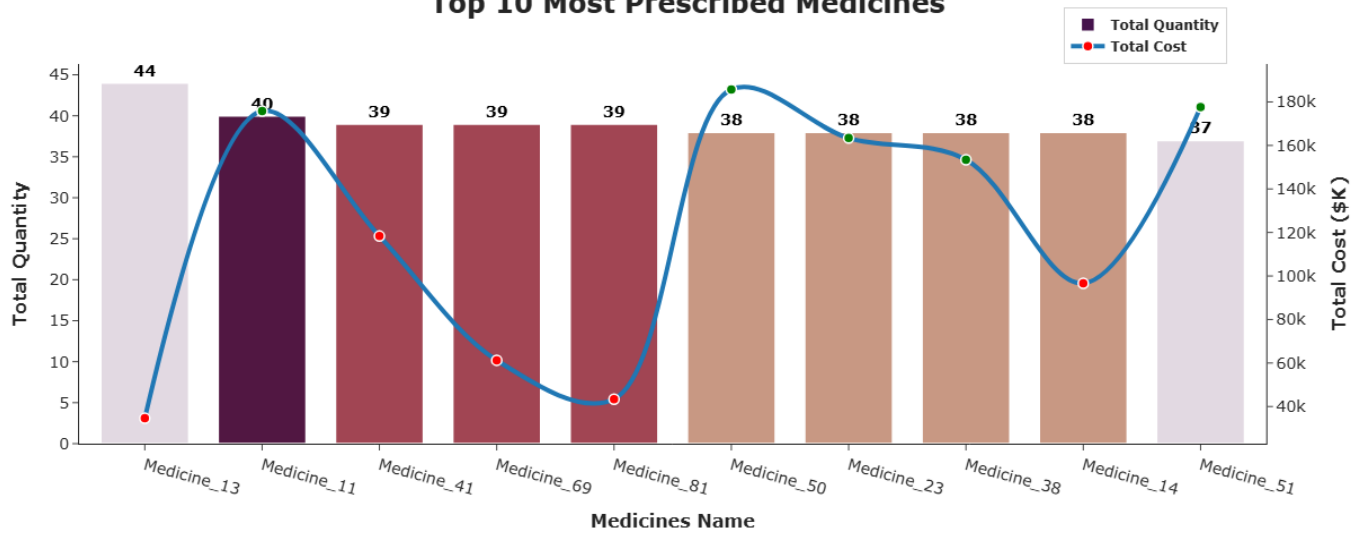


DISEASE & SEVERITY ANALYSIS



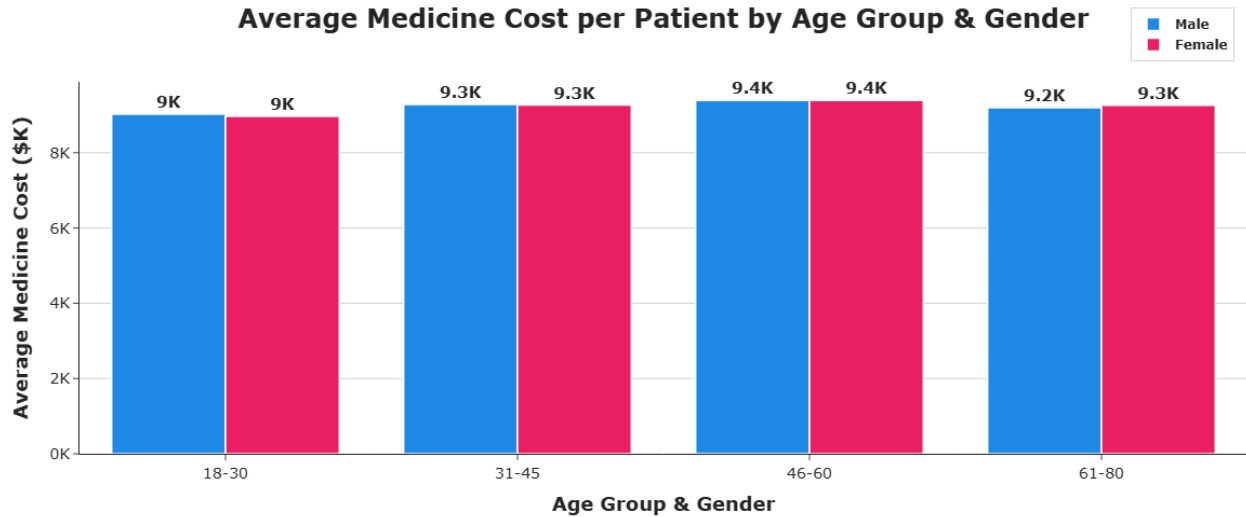
TREATMENT & COST INSIGHTS

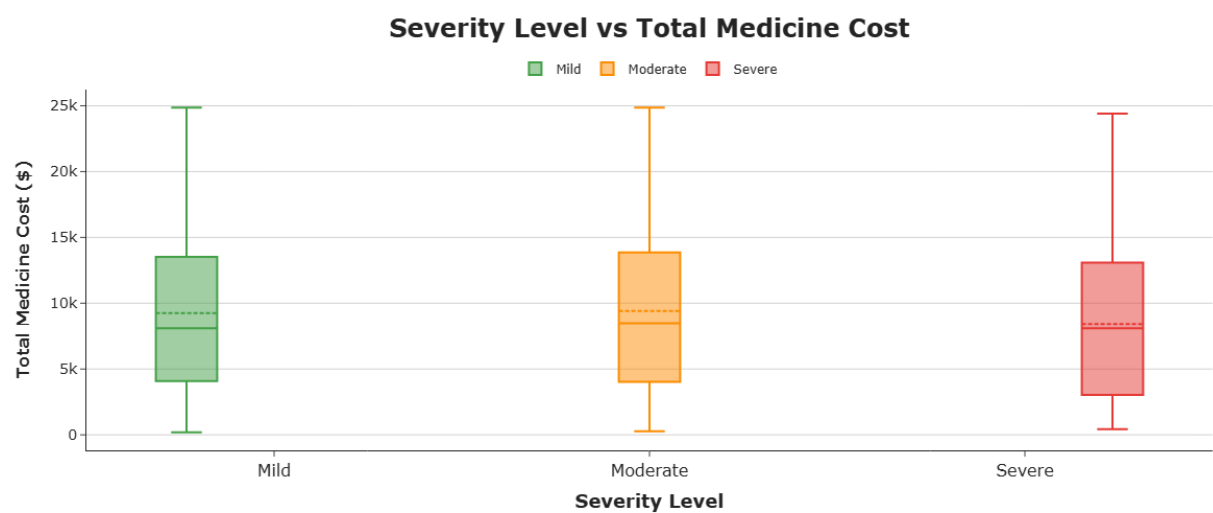
Top 10 Most Prescribed Medicines



Age Group	Gender	Avg Cost	Total Patients
18-30	Female	\$8,969	1
18-30	Male	\$9,027	9
31-45	Female	\$9,266	7
31-45	Male	\$9,285	5
46-60	Female	\$9,390	5
46-60	Male	\$9,389	8
61-80	Female	\$9,258	4
61-80	Male	\$9,200	10

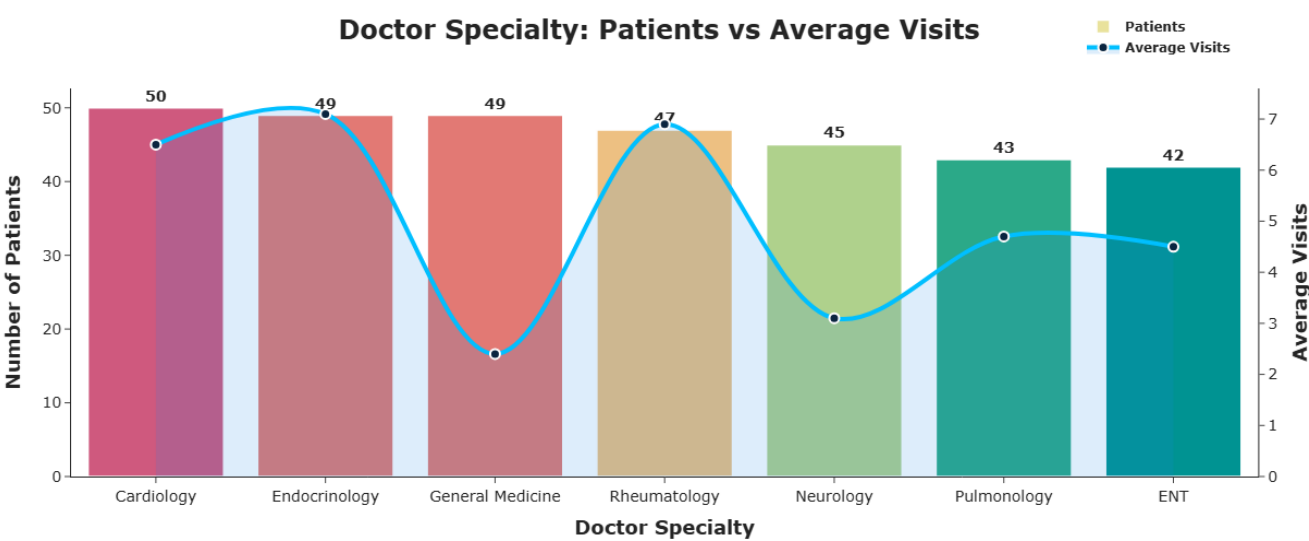
Average Medicine Cost per Patient by Age Group & Gender



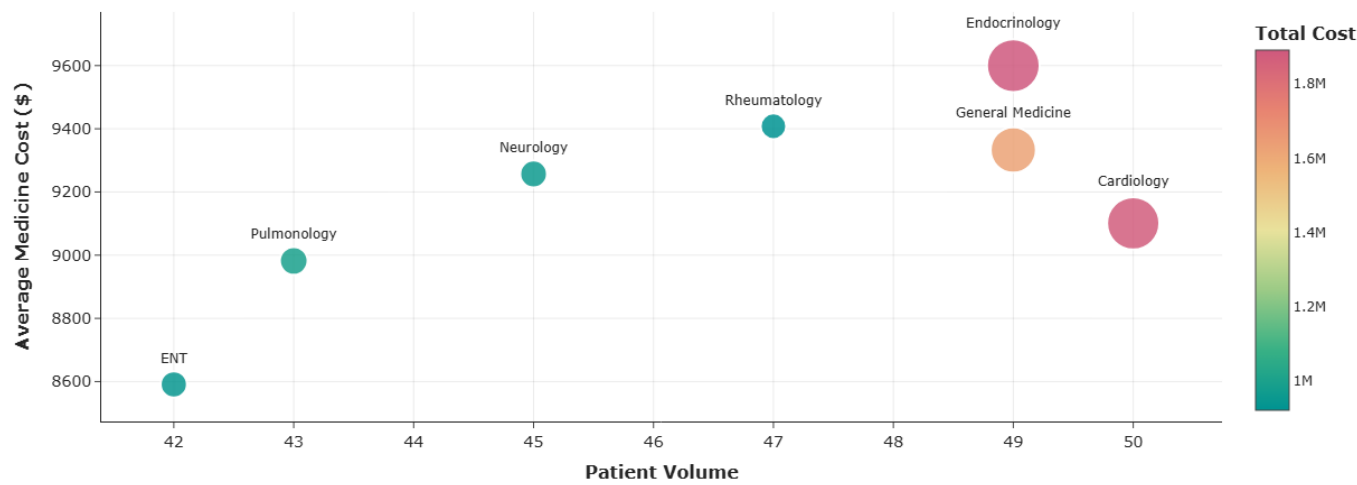


DOCTOR & HOSPITAL PERFORMANCE

Doctor Specialty	Number of Patients	Average Visits
Cardiology	50	6.5
Endocrinology	49	7.1
General Medicine	49	2.4
Rheumatology	47	6.9
Neurology	45	3.1
Pulmonology	43	4.7
ENT	42	4.5

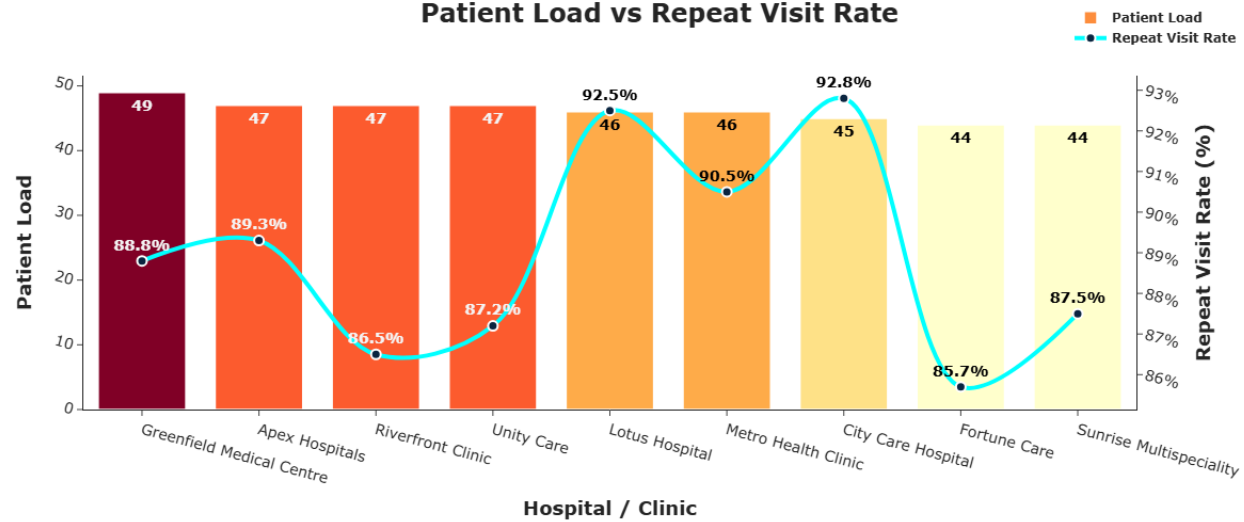


Top Performing vs High-Cost Specialties



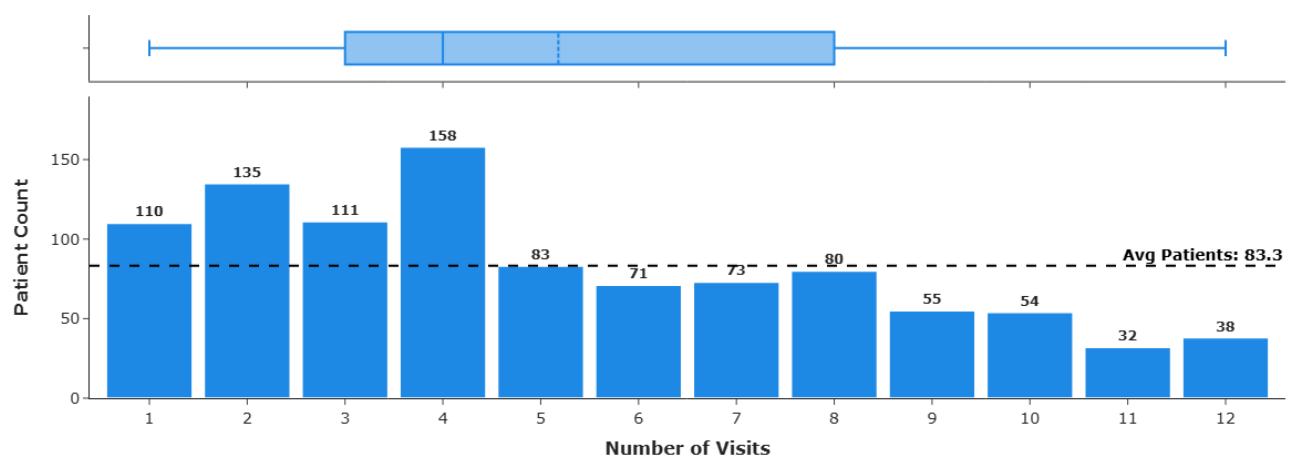
Hospital/Clinic	Number of Patients	Repeat visit Rate	Avg Visits
Greenfield Medical Centre	49	88.8	5.4
Apex Hospitals	47	89.3	5.0
Riverfront Clinic	47	86.5	5.4
Unity Care	47	87.2	4.8
Lotus Hospital	46	92.5	5.1
Metro Health Clinic	46	90.5	5.0
City Care Hospital	45	92.8	5.6
Fortune Care	44	85.7	4.9
Sunrise Multispeciality	44	87.5	5.4

Patient Load vs Repeat Visit Rate



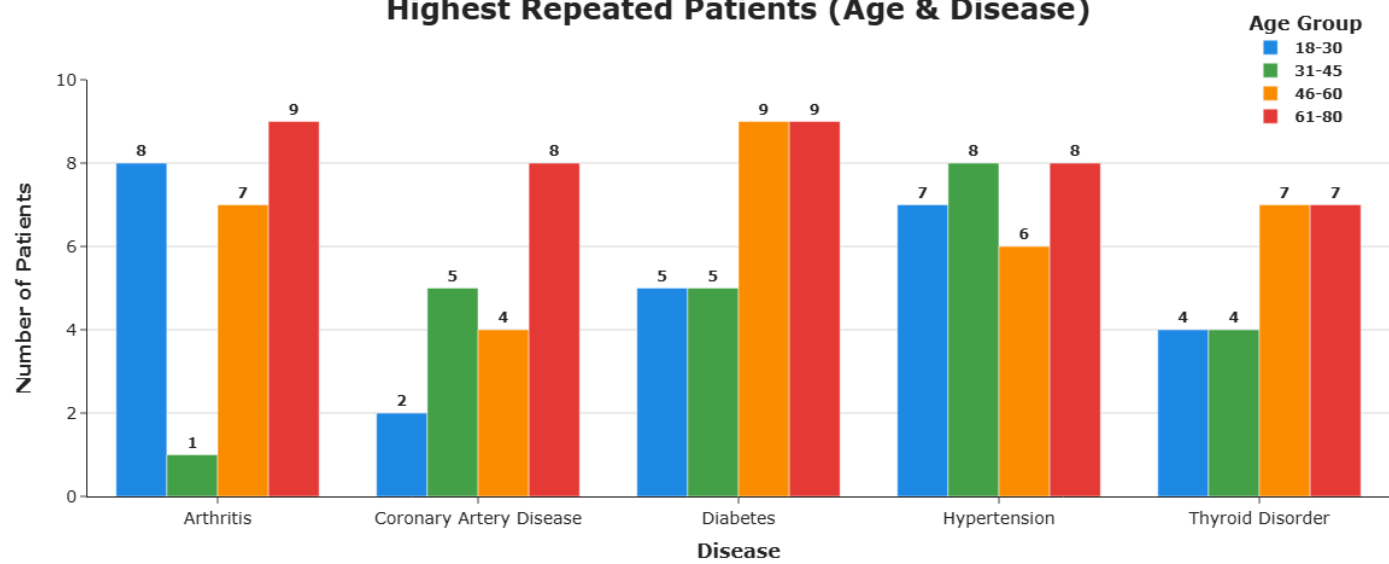
VISIT PATTERNS & EFFICIENCY

Distribution of Patient Visits



	age	severity_num	number_of_visits
age	1.00	0.04	0.05
severity_num	0.04	1.00	0.02
number_of_visits	0.05	0.02	1.00

Highest Repeated Patients (Age & Disease)



BUSINESS RECOMMENDATIONS(SUGGESTIONS)

STRENGTHEN CHRONIC DISEASE MANAGEMENT PROGRAMS

Chronic diseases account for the highest patient volume and treatment expenditure.

Healthcare providers should invest in:

- Preventive care programs
- Regular monitoring systems
- Patient education initiatives
- Digital follow-up services

This can reduce long-term treatment costs and improve patient outcomes.

FOCUS ON HIGH-COST PATIENT SEGMENTS

The analysis shows that **Chronic-Mild** and **Chronic-Moderate** patient groups contribute the largest medicine expenditures.

Recommendations:

- Implement medication optimization programs.
- Review prescription patterns regularly.
- Negotiate better pricing with pharmaceutical suppliers.
- Introduce personalized treatment plans for high-cost patients.

IMPROVE RESOURCE ALLOCATION FOR HIGH-DEMAND SPECIALTIES

Specialties such as:

- Endocrinology
- Rheumatology
- Cardiology

Show higher patient volumes and visit frequencies.

Healthcare organizations should:

- Allocate additional staff and resources.
- Optimize appointment scheduling.
- Expand specialist availability during peak demand periods.

STRENGTHEN PATIENT RETENTION AND FOLLOW-UP STRATEGIES

Repeat visit rates exceed **90%** in several disease categories, particularly chronic conditions.

Suggested actions:

- Develop patient engagement programs.
- Use automated appointment reminders.
- Introduce telemedicine consultations.
- Create long-term care plans for recurring patients.

DEVELOP TARGETED CARE PROGRAMS FOR FREQUENT VISITORS

Patients with more than **8 annual visits** are concentrated in diseases such as:

- Arthritis
- Diabetes
- Hypertension
- Coronary Artery Disease

Specialized disease management programs should be developed to improve treatment effectiveness and reduce unnecessary repeat visits.